

## Cross-Jurisdiction Prediction of Privacy Violation Type and Penalty Severity from Enforcement-Decision Corpora (US + EU)

Problem: Building a machine-learning system that reads real privacy enforcement decisions — EU GDPR fines and **US FTC and California (CCPA/CPPA) actions** — and predicts two things from the text: **which type of violation or legal provision was at issue, and how severe the penalty was.**

**EU GDPR – GDPRhub scraped, 2417 DPA decisions with community written decision texts**

- **GDPR enforcement tracker lacks decision texts entirely, but has “Sector”, to be used for unified schema**

**US FTC – scraped from FTC Legal Library directly, 337 FTC cases, 333 with press-release text which serves as decision text**

- **Rafal’s portfolio, JSON 274 datapoints**
- **Need to move away from Rafal entirely, privated github, dataset not replicable.**
  - **Sounds like a bigger task than it is; Rafal has served as good inspiration and blueprint but our scrape from FTC directly is robust enough on its own.**

**Unified schema: 2,754 cases: 337 US FTC + 2,417 EU GDPRhub**

**Baselines run on unified corpus (nb12). Violation type in-dist macro F1 0.75, severity tier 0.55. Transfer collapses: EU→US 0.23 / US→EU 0.05 (type), EU→US 0.56 / US→EU 0.13 (severity). US→EU degenerates to single-class prediction.**

Motivation: Privacy enforcement is intensifying and splintering across the EU, the FTC, and a fast-growing set of US state laws, and **recent penalties increasingly target subtle failures like dark patterns, unreadable notices, and broken opt-out mechanisms. Tracking all of this by hand doesn't scale, so a model that learns the patterns regulators actually punish has real value for compliance teams, regulators, and researchers.**

Intellectual merit: The only close prior study (**Ruohonen & Hjerppe, 2020**) modeled only EU fines across about 150 cases, using simple TF-IDF features, and found, counterintuitively, that case metadata predicted outcomes better than the decision text. **Questions to answer (1) Do enforcement signals transfer across legal systems — can a model trained on EU decisions classify US actions? (2) Does modern NLP, with transformer models like Legal-BERT, overturn that "metadata beats text" result? Neither has been answered.**

**\*Ruohonen was built on a small dataset that used google translated decision texts, with the GDPR dataset we have built now, we could follow their same pipeline to address:**

- **RQ2 Does modern NLP, with transformer models like Legal-BERT, overturn that "metadata beats text" result?**
- **Bolster metadata to match ruohonen**

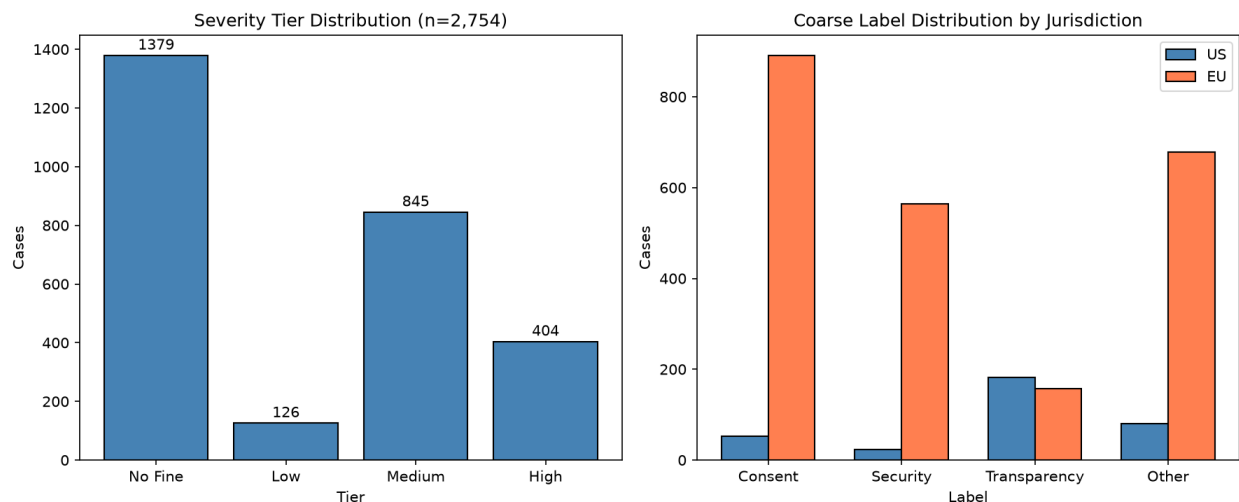
**Deliverable:** The work, in a focused four-week sprint:

**Assemble a harmonized US+EU enforcement dataset from public sources, design a shared violation taxonomy, train classical baselines and then transformers, and run a cross-jurisdiction transfer experiment. A reusable dataset and codebase,** trained models with a proper evaluation, and a short empirical paper — plus hands-on experience in legal-NLP, transformer fine-tuning, cross-domain transfer, messy real-world text, and responsible ML at the law–technology boundary.

**Assemble a harmonized US+EU enforcement dataset from public sources**

**design a shared violation taxonomy**

**Shared violation taxonomy is simple but exists for purpose of ML**



**run a cross-jurisdiction transfer experiment**

**Cross jurisdiction run**

**Overarching RQ:** Do privacy-enforcement signals generalize across legal regimes, and does the contextual-embedding era overturn the standing finding that case metadata predicts outcomes better than the decision text itself?

**Objective.** Given the text and metadata of a privacy enforcement action (an EU DPA decision or a US FTC/CCPA action), predict (a) the violation type(s)/article(s) cited — multi-label classification — and (b) an ordinal penalty-severity tier. A harmonized cross-regime label taxonomy makes EU and US cases jointly modelable and lets the study test whether enforcement signals transfer across legal regimes.

Violation type taxonomy, defined subjectively or objectively?

Penalty severity, via K-means? Find optimal number of clusters using elbow method then apply k means

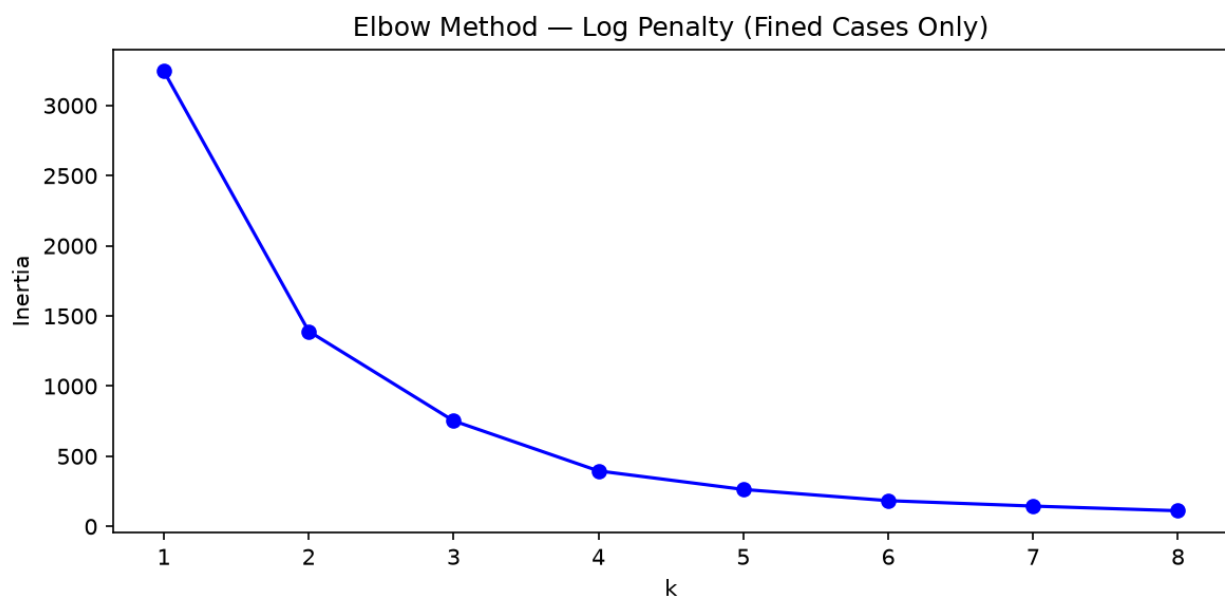
**Week 1 — Acquisition and harmonization.** Assemble the EU corpus (GDPR Enforcement Tracker + GDPRhub decision summaries) and the US corpus (FTC legal library + the prebuilt FTC JSON dataset, California OAG CCPA case examples, and CPPA enforcement actions). Build one unified case schema (jurisdiction, regulator, date, **sector**, statute/article, penalty amount, decision text) and a shared violation-type taxonomy mapping GDPR articles, FTC statutes (Section 5, COPPA, GLBA, FCRA), and CCPA/CPRA sections onto a **common code set** aligned with your existing **95-code framework (disclosure, lawful basis/consent, third-party sharing, security, opt-out/rights, minimization)**. Define the two label targets and run descriptive EDA (label frequencies, penalty distribution, the long-tail/outlier structure of fines).

EU Corpus assembled, need to join “Sector” from GDPR enforcement tracker with GDPRhub cleaned Corpus

Still need “Sector” from GDPR tracker

Preliminary EDA run on scraped GDPRtracker and scraped FTC datasets, need to redo EDA on new more robust datasets

EDA run



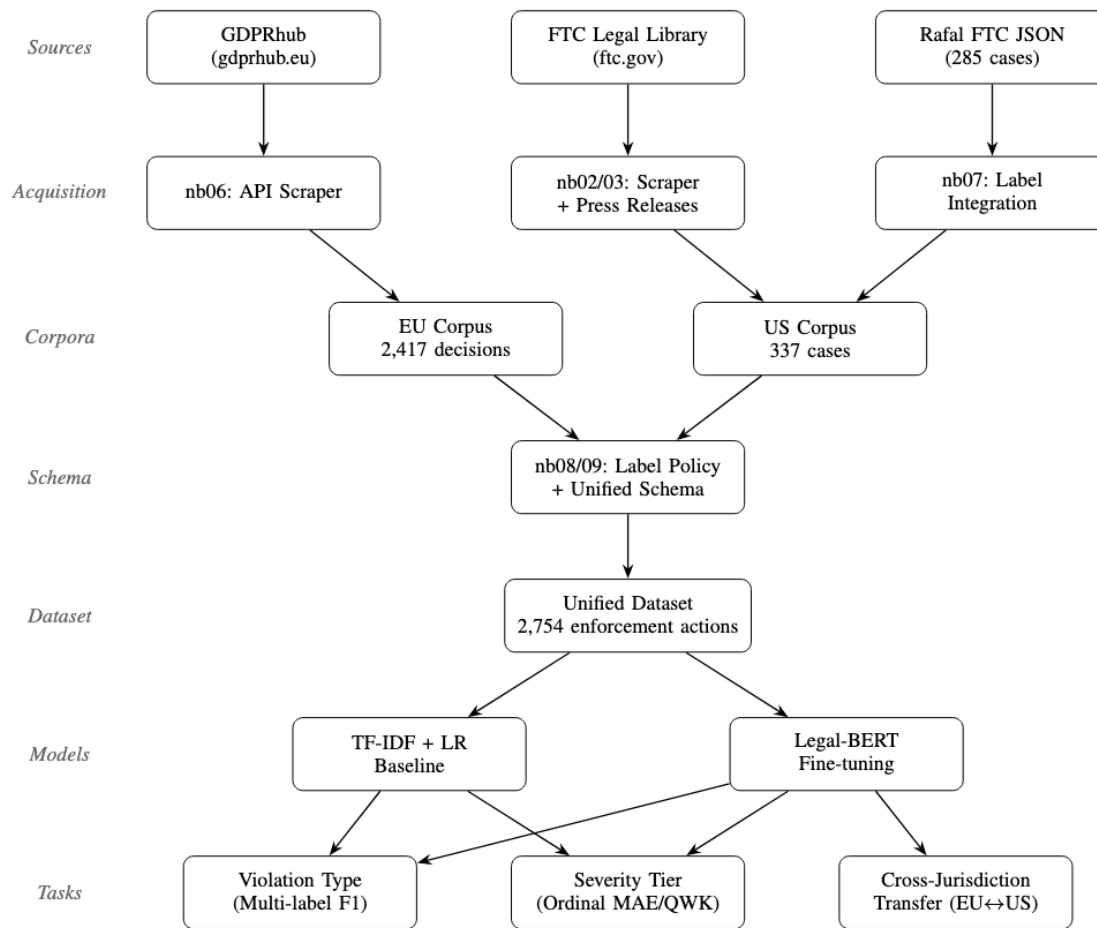


Fig. 1. End-to-end pipeline from raw enforcement data sources to prediction tasks.

**Week 2 — Feature engineering and baselines.** Preprocess text; build TF-IDF plus structured metadata features (regulator, country/state, sector, year). Train baseline models — regularized logistic regression, linear SVM, and gradient boosting — for both the multi-label and the ordinal-tier targets. Stand up the evaluation harness: stratified cross-validation, macro-F1 and per-class F1, AUC, ordinal MAE / quadratic-weighted kappa, and calibration curves. This replicates the established finding that simple models plus metadata are a strong baseline before adding deep models.

TF-IDF and legalBERT preliminary notebook built, this is the biggest area to revisit now that Cross-jurisdiction corpus is assembled

We now have “decision text”, or rather, a surrogate for the decision text for 333 US FTC datapoints, and 2417 EU GDPR datapoints

- For US FTC cases we have:
  - Press release for 337 cases, analyze these 63 and map accordingly
  - We may want to truncate to the 274 cases from Rafal, drop the additional 63 cases that lack Rafal’s mined attributes
  - We have URL match for all 274 of Rafal’s cases
  - Use Rafal as blueprint, reconstruct
- From GDPRhub, we have:
  - Facts - what happened: the background, the complaint, what the company did or failed to do
  - Holding - what the DPA decided: the legal finding, which articles were violated, what remedy or fine was imposed
  - For decision text attribute, Facts and Holding are concatenated (maybe try facts only, holding only, holding + facts?)

**Week 3 — Transformer models and cross-jurisdiction transfer.** Fine-tune Legal-BERT (and optionally a privacy-domain BERT) on the same two tasks; compare against the Week-2 baselines. Address class imbalance and the long tail (resampling, class weighting / focal loss, threshold tuning). Run the transfer experiment the harmonized taxonomy enables: train on EU and test on US (and the reverse), measuring whether a model learned on GDPR decisions generalizes to FTC/CCPA actions.

TF-IDF done for

Fine tuning legalBERT to be done

13 label disagreements from the COPPA/GLBA/FCRA audit in nb08 — cases where Rafal's statutory\_topics and the FTC's own topic tags disagreed. Plus the 3 suspected URL misassignments in Rafal's data (Genica, 214 Technologies, AYLO pointing to wrong FTC pages). 17 cases total to adjudicate by hand.

Audited and corrected

**Week 4 — Analysis, validity framing, write-up.** Error analysis and ablations (metadata-only vs text-only vs combined; EU-only vs US-only vs pooled). Report per-class error rates and calibration explicitly as evidentiary-validity evidence, mapping to the Daubert factors (testable, known error rate, peer-reviewable) and the Frye general-acceptance counterpart already in your framework. State limitations (selection bias toward caught/published cases; many US CCPA "cure" examples carry no monetary label; jurisdictional confounds; mega-fine outliers). Deliver a short empirical paper, a reproducible repository, and a presentation.

In distribution for violation type and severity tier, strong F1 scores

Cross jurisdiction – very bad f1

Train EU test US significantly better than other way around

Thus far, text beats metadata. Our metadata could be more robust, may change finding

Confirmed in paper: text 0.55 vs metadata 0.46, combined 0.56. Low tier is the exception, jurisdiction predicts small fines better than text. Metadata still just jurisdiction + year, article dummies remain the enhancement.

Write up in progress

Findings still needed in writeup, finish previous work etc.

**Full draft complete 07/29. Every section has content: Results with 4 verified tables, Discussion w/ Daubert paragraph, Conclusion, Abstract, all Related Work subsections. Found and fixed a mismatch where Methodology prose misdescribed the taxonomy mapping vs the nb11 code (code was always right, prose corrected).**

Report per-class error rates

mapping to the Daubert factors

testability, peer review, known error rate, standards controlling the technique, and general acceptance.

Frye general-acceptance counterpart

Conceptual framework flow chart, future directions

## Datasets

### European Union (GDPR):

1. **CMS GDPR Enforcement Tracker** — [enforcementtracker.com](https://enforcementtracker.com). The primary EU label source: about 3,194 tracked EU/EEA enforcement actions, updated hourly, each with country, DPA, date, fine amount, sector, violation type, and GDPR article. Scrapeable; the CMS Enforcement Tracker Report 2026 provides aggregate statistics for validation.
2. **GDPRhub** — [gdprhub.eu](https://gdprhub.eu) (operated by noyb). A free, Creative-Commons, open wiki of English-language summaries of national DPA and court decisions, searchable by GDPR article, issuing DPA, or deciding court — richer decision text to complement the tracker's structured fields. [NOYB](https://noyb.eu)

United States (FTC + state privacy law):

1. FTC Legal Library — Cases and Proceedings: <https://www.ftc.gov/legal-library/browse/cases-proceedings> — the official record of cases brought in federal court or through the FTC's internal adjudicative process, covering fraud, identity theft, privacy violations, and more.
2. [Privacy and Security Enforcement | Federal Trade Commission](#)
3. [FTC Act, Data Breach, Data Privacy | JD Supra](#)
4. [FTC Enforcement Provisions Library | Rafal's Portfolio](#)
5. [CCPA Enforcement Case Examples | State of California - Department of Justice - Office of the Attorney General](#)

Auxiliary model/feature resources (optional):

7. **Privacy-policy corpora** (OPP-115, APP-350, PrivacyQA at [usableprivacy.org](https://usableprivacy.org)) for domain-adaptive pretraining or transfer features.
8. **Pretrained models:** Legal-BERT and a privacy-domain BERT.

Need writeup of legal language and definitions as section in paper

Comments for cells in important notebooks

The significance of this study is fourfold. First, it is the first to **harmonize US and EU enforcement under a shared violation taxonomy** and to test whether enforcement signals transfer across legal regimes — a question no prior work poses. Second, it **refreshes a stale precedent** with transformer methods and a substantially larger, current corpus. Third, by reporting **calibration and per-class error rates as Daubert/Frye admissibility evidence**, it directly addresses the methodological validity constraint that legal-ML must satisfy to be credible — a concern your framework already encodes. Fourth, the outputs **feed back into compliance auditing**: a model that learns which textual and structural patterns correlate with enforced violation types becomes a triage tool that can be applied to the policy corpus your existing pipeline already scores, closing the loop between predicted risk and observed enforcement.

1. **harmonize US and EU enforcement under a shared violation taxonomy**
  - a. **Currency normalization**
2. **refresh a stale precedent with transformer methods and a substantially larger, current corpus**
  - a. **LegalBERT, classification**
3. **calibration and per-class error rates as Daubert/Frye admissibility evidence**
  - a. **Daubert, legal parameters**
4. **feed back into compliance auditing\*\***

\*\*Write guide on reproducing dataset due to size

READMEs added to both repos 07/29, cross-linked. MM2 README has pipeline map (live vs superseded notebooks), headline results, prioritized open tasks. Paper README has Overleaf sync workflow + compile notes.

Should be as simple as tweaking file paths, but id like to condense the notebooks, some are extraneous, many are vague

#### Base papers:

1. Albina Orlando, Mario Santoro, *A semantic approach to understanding GDPR fines: From text to compliance insights*, Computer Law & Security Review, Volume 59, 2025, 106187, ISSN 2212-473X, <https://doi.org/10.1016/j.clsr.2025.106187>. (<https://www.sciencedirect.com/science/article/pii/S2212473X25000598>)
2. Ruohonen, J., & Hjerpe, K. (2020/2021). *Predicting the Amount of GDPR Fines*. arXiv:2003.05151; *Information Systems* (Elsevier).

#### References:

3. Wilson, S., Schaub, F., et al. (2016). *The Creation and Analysis of a Website Privacy Policy Corpus (OPP-115)*. ACL. [usableprivacy.org/data](https://usableprivacy.org/data)
4. *MAPS: Scaling Privacy Compliance Analysis to a Million Apps*. Sebastian Zimmeck, Peter Story, Daniel Smullen, Abhilasha Ravichander, Ziqi Wang, Joel Reidenberg, N. Cameron Russell, and Norman Sadeh. Privacy Enhancing Technologies Symposium 2019.
5. Abhilasha Ravichander, Alan W Black, Shomir Wilson, Thomas Norton, and Norman Sadeh. 2019. *Question Answering for Privacy Policies: Combining Computational and Legal Perspectives*. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 4947–4958, Hong Kong, China. Association for Computational Linguistics.
6. Wasi Ahmad, Jianfeng Chi, Yuan Tian, and Kai-Wei Chang. 2020. *PolicyQA: A Reading Comprehension Dataset for Privacy Policies*. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 743–749, Online. Association for Computational Linguistics.
7. J. Cui, X. Shen and S. Wen, "A Survey on Legal Judgment Prediction: Datasets, Metrics, Models and Challenges," in *IEEE Access*, vol. 11, pp. 102050-102071, 2023, doi: 10.1109/ACCESS.2023.3317083.
8. Adhikari, A., Das, S., & Dewri, R. (2025). *Natural Language Processing of Privacy Policies: A Survey*. ArXiv. <https://arxiv.org/abs/2501.10319> *Natural Language Processing of Privacy Policies: A Survey*



9. [Rui Zhao, Vladyslav Melnychuk, Jun Zhao, Jesse Wright, Nigel Shadbolt: \*An LLM-enabled semantic-centric framework to consume privacy policies\*. \*CoRR abs/2509.01716\* \(2025\)](#)
10. Zhao, R., Melnychuk, V., Zhao, J., Wright, J., & Shadbolt, N. (2025). *Let's Measure the Elephant in the Room: Facilitating Personalized Automated Analysis of Privacy Policies at Scale*. ArXiv. <https://arxiv.org/abs/2507.14214>**APPSI-139**: A Parallel Corpus of English Application Privacy Policy Summarization and Interpretation (2026). arXiv:2604.27550.
11. [\*Predictive Modelling in Legal Decision-Making: Forecasting Legal Outcomes\* \(2024\), incl. DPDPA online-privacy verdict prediction. ResearchGate 380981542.](#)
12. CMS Law (2026). *GDPR Enforcement Tracker Report 2025/2026*. [GDPR and beyond: The impact of initial sanctions and new regulations](#)
13. [Privacy Platform for the GDPR | DataGrail](#)
14. IAPP (2024). *FTC Enforcement Trends: From Straightforward Actions to Technical Allegations*. [iapp.org/resources/article/ftc-enforcement-trends](https://iapp.org/resources/article/ftc-enforcement-trends)
15. California Office of the Attorney General. *CCPA Enforcement Case Examples*. [oag.ca.gov/privacy/ccpa/enforcement](https://oag.ca.gov/privacy/ccpa/enforcement).
16. [Koley Jessen / Troutman \(2026\). \*California AG Announces Largest CCPA Settlement to Date\* \(Disney/ABC, \\$2.75M\); CPPA actions \(Ford, PlayOn Sports\).](#)
17. [GDPR Compliance in 2026: The Complete Guide | Secure Privacy Blog](#)
18. [GDPR Fines Hit €7.1 Billion: Data Privacy Enforcement Trends in 2026](#)

**What needs to be done immediately:**

**LegalBERT fine tuning**

**Sector join from GDPR enforcement tracker**

**Notebook comments so whoever takes over can understand what has happened thus far**

**Finish rough draft of paper**

**Notebook modularity, relative file paths?**

**Do binary classification first**

**Combine low-medium-high into one tier, normalized no fine or fine**

**Status 07/29 (graduated, handoff mode):**

**| DONE: notebook comments, rough draft, repro guide (READMEs)**

**| OPEN, priority order: LegalBERT fine-tuning (starter code lives in nb05), sector join from tracker (nb01 has the scrape), build out metadata (article dummies), binary fine/no-fine classification, notebook modularity/relative paths, CCPA**

**| Known quirk: nb12 Task 2 cell prints 0.52 (seed 777) vs paper's 0.55 (seed 42, Ruohonen cell). Documented in nb12. Regenerate from one cell consistently.**

## **BIG PICTURE**

**California/CCPA**

**Decision text, is there an access angle im not considering?**

**Build out metadata**